ITTA Special HAZMAT Publication 800-001A

Guideline for using HAZMAT respirator regulation 800-001A

v1.4.0

Authority

This publication for the Ruby Mountains Superstructure has been developed by the ITAA in coordinance with the SRA. This guideline is responsible for ensuring proper production and safety standards, including minimum requirements for such equipment, however, this standard may be overridden given proper clearance to do so.
Nothing in this publication shall be released to the general public, as it is confidential.

Institution for Technological Action and Advocacy
Special HAZMAT Publication 800-001A

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the Board of Directors
The Administrator (with permission)

Abstract

This regulation document is intended to serve hazmat units within the Superstructure, and to adapt to newer technologies now available, along with increased relative safety standards. This Special Publication (SP) is the first one of its kind, but will be iterated on.

Keywords

confidentiality; critical infrastructure; Administrative directives; mandates; policy; standards.

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Contents

1 INTRODUCTION

1.1 Background and Purpose

HAZMAT publications of the Institute for Technological Action and Advocacy (ITAA) provide guidance regarding equipment usage, manufacturing requirements, situation control, and what to use in said situations.
This document provides basic guidance on how to identify what filter should be used in HAZMAT situations where SCBA equipment is either unavailable or not required, along with other information regarding filter construction, proper usage, and traits.

1.2 Acronyms

ITAA	Institution for Technological Action and Advocacy
SRA	Superstructure Research Authority
BoD	Board of Directors
HAZMAT	Hazardous-Materials
SCBA	Self-Contained Breathing Apparatus
ppm	parts per million
SP	Special Publication
v/v	volume per volume
IS	Inserts Chart
IEC	Insert Efficiency Chart
SD	Short Duration
MD	Medium Duration
HD	High Duration
XHD	Extra-High Duration
Hi	High
Lo	Low

1.3 Document Organization

- Section 1 provides an introduction to this document, including its background and purpose, and a list of acronyms used herein.
- Section 2 provides basic filter identification.
- Section 3 describes information on what should be the base main filter types, the size standards of said filter types, and accepted filter casing materials.
- Section 4 provides other information.

2 Filter Identification

This section provides basic filter identification, along with a list of the main filter types.

2.1 Filter Trait Charts

2.1.1 Inserts Chart

Each insert can be applied into a filter, be it one or more.

Types	Traits	Colors
AX	Gases and vapours of organic compounds with a boiling point below 65 degrees.	
A	Gases and vapours of organic compounds with a boiling point above 65 degrees.	
B	Inorganic gases & vapours[1].	
E	Sulphur dioxide & hydrogen chloride.	
K	Ammonia and its organic derivatives.	
CO	Carbon monoxide.	
Hg	Mercury vapour.	
NO	Nitrous gases, including nitrogen monoxide.	
RK	Radioactive iodine, including radioactive methyl iodine.	
P	Particulates.[2]	

2.1.2 Insert Efficiency Chart

Filtration efficiency ranges from 1-5, 5 being the best and 1 the worst.
Each filter insert can have its own filtering efficiency,
be it to reduce complexity / cost or conserve space.

Types	Reduction (ppm)	Reduction (Particulate, ppm)
C5	90,000 ppm (9% v/v)	999,999 ppm (99.9999% v/v) - (log-6)
C4	50,000 ppm (5% v/v)	900,000 ppm (90% v/v) - (log-1)
C3	10,000 ppm (1% v/v)	800,000 ppm (80% v/v) - (log-0.7)
C2	5,000 ppm (0.5% v/v)	700,000 ppm (70% v/v) - (log-0.53)
C1	1,000 ppm (0.1% v/v)	600,000 ppm (60% v/v) - (log-0.4)

[1]: Chlorine, Hydrogen sulphide, etc.
[2]: Particulate filter reduction amount ranges from 5 - $\binom{99.9999\%}{log-6red.}$ to 1 - $\binom{60\%}{0.4}$

3 Main filter types

3.1 Main Filter Types

3.1.1 Main Filter Duration Chart*

Types	Durations	Avg. Durations
SD	1H(High-exposure) - 8H(Low-exposure)	4.5 Hours
MD	4H(High-exposure) - 18H(Low-exposure)	11 Hours
HD	8H(High-exposure) - 30H(Low-exposure)	19 Hours
XHD	12H(High-exposure) - 60H(Low-exposure)	36 Hours

3.1.2 Main Filter Type Chart

Types	Traits	Types
Omni(Combined)	Must contain all inserts.	Hi/Lo
REACTOR	Must contain RK-type insert.	Hi/Lo
Particulate	P-type and B-type inserts only.	Hi/Lo
Vapour	May contain all inserts excluding RK-type.	Hi/Lo

*Do note that these are the expected durations, and can be exceeded, however if the design falls short of these standards, the BoD may be advised in order to test if the filter is still sufficient for use.

3.2 Main Filter Parameters

3.2.1 Sizes & Casings Chart

Types	Sizes	Accpeted Casing Materials
Hi-Omni[SD]	12(cm)x10(cm)	Nomex / Titanium Blend
Lo-Omni[SD]	10(cm)x10(cm)	Nomex / Titanium Blend
Hi-Omni[MD]	12(cm)x15(cm)	Nomex / Titanium Blend
Lo-Omni[MD]	10(cm)x15(cm)	Nomex / Titanium Blend
Hi-Omni[HD]	12(cm)x20(cm)	Nomex / Titanium Blend
Lo-Omni[HD]	10(cm)x20(cm)	Nomex / Titanium Blend
Hi-Omni[XHD]	12(cm)x30(cm)	Nomex / Titanium Blend
Lo-Omni[XHD]	10(cm)x30(cm)	Nomex / Titanium Blend
Hi-REACTOR[MD]	12(cm)x10(cm)	Nomex / Titanium Blend
Lo-REACTOR[MD]	10(cm)x10(cm)	Nomex / Titanium Blend
Hi-REACTOR[HD]	12(cm)x15(cm)	Nomex / Titanium Blend
Lo-REACTOR[HD]	10(cm)x15(cm)	Nomex / Titanium Blend
Hi-REACTOR[XHD]	12(cm)x25(cm)	Nomex / Titanium Blend
Lo-REACTOR[XHD]	10(cm)x25(cm)	Nomex / Titanium Blend
Hi-Particulate[MD]	12(cm)x5(cm)	Nomex
Lo-Particulate[MD]	10(cm)x5(cm)	Nomex
Hi-Particulate[HD]	12(cm)x10(cm)	Nomex
Lo-Particulate[HD]	10(cm)x10(cm)	Nomex
Hi-Particulate[XHD]	12(cm)x15(cm)	Nomex
Lo-Particulate[XHD]	10(cm)x15(cm)	Nomex
Hi-Vapour[SD]	12(cm)x5(cm)	Nomex / Titanium Blend
Lo-Vapour[SD]	10(cm)x5(cm)	Nomex / Titanium Blend
Hi-Vapour[MD]	12(cm)x10(cm)	Nomex / Titanium Blend
Lo-Vapour[MD]	10(cm)x10(cm)	Nomex / Titanium Blend
Hi-Vapour[HD]	12(cm)x15(cm)	Nomex / Titanium Blend
Lo-Vapour[HD]	10(cm)x15(cm)	Nomex / Titanium Blend
Hi-Vapour[XHD]	12(cm)x25(cm)	Nomex / Titanium Blend
Lo-Vapour[XHD]	10(cm)x25(cm)	Nomex / Titanium Blend

Format is as following: Radius(cm/mm) x length(cm).
Other filter casings can be discussed with the BoD.

4 Other

4.1 Important

4.1.1 Main

4.1.2 Sources

https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-175A.pdf (helped in document construction, also borrowed some text to reformat)
https://whiteknuckle.miraheze.org/wiki/White_Knuckle_Wiki (bery goodies game, go play it, this is a fan-document for wuckle anyways)